



Innovation & Entrepreneurship Hub for Educated Rural Youth SURE ProEd (formerly SURE Trust)

RiskCheck AI

Invoice Fraud Detection & Investigation Prioritisation

The domain of the Project:

DATA ANALYTICS

GROUP -1

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Declaration

The project titled **RiskCheck AI: Invoice Fraud Detection & Investigation Prioritisation** has been mentored by **Ms. Nithyasri Kannathal EL** ma'am and **Mr. Bhargavesh Dakka** sir, organised by SURE Trust, from February 2026 to September 2026 for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

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Executive Summary

RiskCheck AI is an end-to-end invoice fraud detection and investigation-prioritization system developed as the final capstone project of my Data Analytics Internship and Capstone Program with SURE ProEd, formerly known as SURE Trust. The project demonstrates how data analytics, machine learning, cloud-based model deployment, and human verification can be combined to support accounts payable teams in detecting suspicious invoices and managing fraud investigations more efficiently.

Organizations regularly process large numbers of invoices received from multiple suppliers. Manually reviewing every invoice can be time-consuming and may not provide a consistent way to identify the transactions that require immediate attention. Suspicious invoices may contain indicators such as unusually high amounts, duplicate invoice details, split transactions, blacklisted suppliers, newly registered suppliers, late-night submissions, unusual payment terms, differences between invoice totals and OCR-extracted values, or signs of image manipulation. These indicators may be difficult to evaluate together through manual review alone.

RiskCheck AI addresses this challenge by analyzing invoice information and estimating the probability of fraud for each transaction. The application accepts multiple invoice files in CSV or Parquet format. Each record must contain an invoice identifier and 32 behavioral, supplier-related, transactional, temporal, financial, and document-related features. The system validates the uploaded data before submitting the required features to a machine learning model deployed through Databricks Model Serving.

The model returns a fraud probability between zero and one for every invoice. RiskCheck AI converts this probability into a score between 0 and 100 and assigns one of five investigation priorities. Priority 1 represents critical-risk invoices requiring immediate investigation. Priority 2 represents high-risk invoices requiring urgent manual review. Priority 3 represents elevated-risk invoices requiring standard review. Priority 4 represents moderate-risk invoices that should be monitored or considered for secondary review. Priority 5 represents low-risk invoices that can continue through routine processing.

This priority-based approach helps investigators focus their time and attention on the invoices with the greatest estimated risk. It improves the consistency of invoice screening and provides a structured investigation queue instead of treating every invoice as equally important.

The system also includes an automated email-notification process. When an invoice is classified as Priority 1 or Priority 2, RiskCheck AI can send a formatted HTML alert to the appropriate accounts payable or fraud-review team. The alert contains invoice-level risk information and a direct link to the relevant batch, helping reviewers access high-priority cases quickly.

To maintain an audit trail, all scored invoices are stored in a Databricks Delta table. Investigators can review flagged invoices and record a verified outcome of either Fraud or Legitimate. These outcomes are stored separately from the model predictions, making it possible to distinguish automated risk estimates from confirmed human decisions.

The verified feedback also supports controlled model improvement. When the number of new and unused investigator labels reaches a configured threshold, the application can trigger an automated Databricks



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retraining job through a REST API. This feedback-driven process enables the model to learn from newly confirmed cases while keeping human judgement at the Centre of the fraud-investigation workflow.

The completed solution uses Python and Streamlit for the application interface, Databricks Model Serving for fraud prediction, Databricks SQL and Delta Lake for persistent storage, SMTP for email notifications, and Databricks Jobs for automated retraining. The application is deployed through Streamlit Community Cloud, and the complete source code is maintained in GitHub.

RiskCheck AI is designed as a decision-support system rather than an autonomous fraud-decision system. A high score indicates that an invoice requires additional investigation; it does not prove that fraud has occurred. The final decision remains with an authorised investigator. By combining automated risk scoring, priority-based review, persistent audit records, human feedback, and controlled retraining, the project presents a practical approach to improving the efficiency and reliability of invoice fraud investigations.

Introduction

1.1 Background

Invoice processing is an important activity within the accounts payable department of an organization. Businesses receive invoices from suppliers for products and services, verify their details, obtain the required approvals, and process payments. In organizations that handle a high volume of invoices, reviewing every transaction manually can require considerable time and effort.

Invoice data may contain several indicators of possible risk. These include unusually high invoice amounts, duplicate invoices, split invoices, blacklisted suppliers, recently registered suppliers, late-night submissions, transactions outside normal business hours, differences between invoice and OCR-extracted amounts, and evidence of image manipulation. Although an individual indicator does not prove fraud, a combination of unusual patterns may require further investigation.

Traditional rule-based systems can identify transactions that meet predefined conditions. However, fixed rules may not recognize complex relationships between supplier behavior, invoice characteristics, historical transaction patterns, and document-related information. Manual review also becomes difficult when investigators must examine many invoices without a consistent method of prioritization.

Machine learning can support this process by analyzing multiple invoice features together and estimating the probability of fraud. The result can be used to rank invoices according to their level of risk. Investigators can then concentrate on high-risk transactions while allowing low-risk invoices to continue through routine processing. RiskCheck AI was developed to demonstrate this risk-based approach. It combines invoice data validation, machine learning, risk scoring, automated alerts, persistent storage, investigator feedback, and model retraining within a single application.



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1.2 Problem Statement

Accounts payable by teams need to process invoices efficiently while protecting organizations from financial loss and reporting errors. When invoice volumes are high, manually reviewing every transaction with the same level of attention is not practical. Investigators require a method for identifying which invoices should be reviewed first.

Several challenges affect invoice-fraud investigation:

- Suspicious activity may be distributed across several invoices and supplier characteristics.
- Duplicate or split invoices may not be immediately visible during manual processing.
- New or high-risk suppliers may require additional verification.
- Differences between invoice values and OCR-extracted values may indicate document errors or manipulation.
- Review decisions may not be stored in a structured form for future analysis.
- Machine learning models may become less effective if they are not updated using newly verified outcomes.
- Fraud predictions must remain explainable, traceable, and subject to human review.

The central problem addressed by RiskCheck AI is therefore the development of a system that can analyze invoice records, estimate fraud risk, prioritize investigations, notify the appropriate reviewers, and learn from verified investigation outcomes.

The application does not automatically declare that an invoice is fraudulent. A high score indicates that the invoice contains patterns associated with greater risk and should receive additional attention. The final decision remains with an authorised investigator.

1.3 Project Scope

RiskCheck AI supports an end-to-end invoice risk-assessment workflow. The project includes:

- Uploading invoice batches in CSV and Parquet formats.
- Validating the invoice identifier and 32 required model features.
- Sending validated records to a Databricks Model Serving endpoint.
- Receiving a fraud probability for each invoice.
- Converting fraud probabilities into RiskCheck Scores from 0 to 100.
- Classifying invoices into five investigation-priority levels.
- Sending automated email alerts for critical-risk and high-risk invoices.
- Storing scored invoice records in a Databricks Delta table.



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- Allowing investigators to classify reviewed invoices as Fraud or Legitimate.
- Storing verified outcomes in a separate feedback table.
- Triggering a Databricks retraining job when enough new verified labels are available.
- Providing an interactive Streamlit interface for uploading, scoring, reviewing, and monitoring invoice batches.

The project focuses on fraud-risk prediction and investigation support. It does not make final legal or financial determinations, automatically block supplier payments, or replace an organization's internal approval procedures. Its purpose is to provide consistent risk information that helps accounts payable teams and fraud investigators make better-informed decisions.

1.2 Problem Statement

Teams accounts payable must process invoices quickly while protecting their organizations from fraud, duplicate payments, incorrect charges, and manipulated documents. In a high-volume environment, manually examining every invoice in detail is inefficient and may delay legitimate supplier payments. It also creates a risk that suspicious invoices will not receive attention at the correct time.

Invoice fraud is difficult to detect because suspicious activity does not always depend on one obvious condition. An invoice may appear normal when considered independently but becomes unusual when compared with the supplier's previous transactions, average invoice amount, business operating hours, payment terms, or department budget.

The main problem addressed by RiskCheck AI is the absence of a consistent and scalable method for evaluating invoice risk and prioritizing investigations. Accounts payable by teams require a system that can:

- Process multiple invoice files efficiently.
- Validate the information required for fraud prediction.
- Evaluate several risk indicators together.
- Assign a consistent risk score to every invoice.
- Direct investigators are looking for the highest risk transactions.
- Notify relevant teams when urgent cases are detected.
- Maintain an audit trail of predictions and review decisions.
- Use verified investigation outcomes to support future model improvement.

RiskCheck AI addresses this problem by combining machine learning predictions with an investigation workflow. The system estimates the risk of each invoice, converts the prediction into an understandable



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score and assigns a priority level. This approach helps investigators decide which invoices should be reviewed first.

1.3 Project Scope

The scope of RiskCheck AI covers the complete process from invoice upload to verified investigator feedback. The system supports the following activities:

- Uploading multiple invoice batches in CSV and Parquet formats.
- Validating invoice_id and the 32 model features required for prediction.
- Preparing invoice records for submission to a Databricks Model Serving endpoint.
- Processing large batches through chunked API requests.
- Receiving probabilities for individual invoices.
- Converting probabilities into RiskCheck Scores between 0 and 100.
- Classifying invoices into five investigation-priority levels.
- Send email alerts for Priority 1 and Priority 2 invoices.
- Storing scored invoice records in a Databricks Delta table.
- Allowing investigators to record Fraud or Legitimate outcomes.
- Storing human-verified outcomes in a separate feedback table.
- Monitoring the number of new labels available for retraining.
- Triggering a Databricks retraining job when the configured threshold is reached.
- Providing a Streamlit interface for scoring and reviewing invoice batches.

The project focuses on invoice risk assessment and investigation support. It does not automatically approve or reject invoices, stop supplier payments, create legal fraud determinations, or replace an organization's existing financial controls.

1.4 Project Limitations

RiskCheck AI has several limitations that must be considered when interpreting its results.

First, the quality of each prediction depends on the quality and relevance of the uploaded invoice data. Missing, incorrectly formatted, or inaccurate features may reduce the reliability of the fraud score.

Second, the model identifies statistical patterns associated with fraud. It cannot independently prove fraudulent intent. A high-risk invoice may be legitimate, while a low-risk invoice may still require investigation if additional information becomes available.



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Third, the application depends on external services, including Databricks Model Serving, Databricks SQL, the email server, and the retraining job. Service availability, access permissions, or configuration errors may affect certain functions.

Fourth, automated retraining depends on investigators providing accurate verified labels. Incorrect feedback could affect the quality of later model versions. Retraining should therefore include appropriate validation and approval procedures before a new model is deployed.

Finally, the current priority boundaries are predefined. Different organizations may require thresholds that reflect their fraud tolerance, investigation capacity, regulatory obligations, and financial exposure.

1.5 Innovation Component

The main innovation of RiskCheck AI is the integration of fraud scoring, investigation prioritization, automated communication, audit storage, human verification, and model retraining in one workflow.

Many machine learning projects end after producing a prediction. RiskCheck AI extends the prediction process by converting model output into operational actions. High-risk invoices generate alerts, reviewers can access the relevant batch, investigation outcomes are stored for audit purposes, and verified labels can later support model retraining.

The complete workflow can be represented as:

Invoice Upload → Data Validation → Fraud Prediction → Risk Prioritization → Reviewer Alert → Human Verification → Model Retraining

This feedback-driven design keeps investigators involved in the final decision while allowing the machine learning model to improve from verified outcomes. It demonstrates a practical and responsible approach to applying artificial intelligence within financial operations.

Project Objectives

2.1 Main Objective

The main objective of RiskCheck AI is to develop an end-to-end invoice fraud detection and investigation-prioritization system that helps accounts payable teams identify suspicious transactions, organize manual reviews, maintain an audit trail, and improve the fraud detection model using verified investigator feedback.

The project aims to transform a machine learning prediction into a practical business workflow. Instead of presenting only a fraud probability, the system converts the prediction into an understandable RiskCheck Score and assigns an investigation priority. This allows reviewers to focus on invoices with the greatest estimated risk.



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2.2 Specific Objectives

The specific objectives of the project are:

1. To analyze invoice fraud risk factors

The project aims to evaluate invoice characteristics associated with potential fraud, including unusual invoice amounts, duplicate invoices, split invoices, supplier history, blacklisting information, submission time, payment terms, OCR differences, and document-integrity indicators.

2. To prepare and validate invoice data

The system must confirm that every uploaded file contains an invoice identifier and the 32 features required by the machine learning model. This validation prevents incomplete or incorrectly structured data from being submitted for prediction.

3. To support multi-file batch processing

The application should allow users to upload multiple CSV or Parquet files and score a large number of invoices during one session. Large datasets should be divided into manageable chunks before being sent to the model-serving endpoint.

4. To deploy the fraud detection model

The trained machine learning model should be made available through Databricks Model Serving. The Streamlit application should communicate with this endpoint through REST API requests and receive a fraud probability for each invoice.

5. To create an understandable risk-scoring method

The returned fraud probability should be converted into a RiskCheck Score between 0 and 100. The score should make the model output easier for accounts payable employees and investigators to interpret.

6. To prioritize invoice investigations

Each invoice should be assigned to one of five investigation priorities: Critical, High, Elevated, Moderate, or Low. This classification should provide a consistent method for deciding which invoices require immediate investigation, and which can continue through routine processing.

7. To automate high-risk notifications

The system should send formatted email alerts when Priority 1 or Priority 2 invoices are detected. The alert should provide important invoice-risk information and a direct link to the relevant review batch.

8. To maintain a persistent audit trail

All scored invoices should be stored on a Databricks Delta table. This will allow the organization to preserve prediction results, batch information, risk scores, and priority classifications for investigation and audit purposes.



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9. To collect verified investigator feedback

Investigators should be able to record whether a reviewed invoice is Fraud or Legitimate. These verified outcomes should be stored separately from model predictions to ensure traceability.

10. To support automated model retraining

The system should monitor the number of new verified labels that have not been used for retraining. When the configured minimum threshold is reached, the application should trigger a Databricks retraining job through a REST API.

11. Developing an interactive application

A Streamlit application should bring together file uploading, validation, scoring, investigation review, feedback collection, and retraining status in a user-friendly interface.

12. To apply responsible machine learning

The system should operate as a decision-support tool. It should help investigators identify potentially suspicious invoices without automatically declaring fraud or replacing human judgement.

2.3 Expected Outcomes and Deliverables

The expected outcomes and deliverables of RiskCheck AI are:

- A prepared invoice fraud dataset containing relevant model features.
- A trained machine learning model for estimating invoice fraud probability.
- A Databricks Model Serving endpoint for real-time batch prediction.
- A Streamlit application for uploading and scoring invoice files.
- Support for CSV and Parquet invoice batches.
- Automated validation of the required upload schema.
- A RiskCheck Score ranging from 0 to 100.
- A five-level investigation-priority framework.
- Automated email alerts for critical-risk and high-risk invoices.
- A Delta table containing scored invoice records.
- A separate Delta table containing verified investigator feedback.
- A feedback interface for recording Fraud and Legitimate outcomes.
- An automated threshold checks for newly verified labels.
- A Databricks retraining job that can be triggered through the application.



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- Sample invoice files for application testing and demonstration.
- A deployed Streamlit application.
- A GitHub repository containing the project source code, notebooks, requirements, and supporting files.

By achieving these objectives, RiskCheck AI provides a complete demonstration of how data analytics and machine learning can be applied to invoice fraud investigation. The project connects technical model development with practical business actions such as risk prioritization, alerts, audit storage, human review, and continuous improvement.

Methodology and Results

3.1 Project Workflow

RiskCheck AI was developed using an end-to-end machine learning and application-development workflow. The process began with understanding the invoice fraud problem and identifying the information required to evaluate invoice risk. The data was then prepared, validated, analyzed, and transformed into model-ready features.

After model development, the trained fraud detection model was deployed through Databricks Model Serving. A Streamlit application was created to upload invoice batches, validate their structure, request predictions, calculate risk scores, and organize invoices according to their investigation priority.

The project workflow can be represented as:

Invoice Data → Data Validation → Feature Engineering → Model Training → Model Serving → Risk Scoring → Investigation Priority → Human Feedback → Model Retraining

This workflow connects the analytical and operational parts of the project. Model predictions are not treated as the result. Instead, they are used to support reviewer actions, store investigation outcomes, and improve future model versions.

3.2 Dataset and Input Structure

The dataset used for the project contains invoice-level information relating to transaction value, supplier behavior, submission timing, payment terms, department budgets, OCR validation, and document integrity.



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Each invoice record contains a unique invoice_id and 32 model features. These features can be grouped as follows:

Feature category	Examples
Invoice value	Invoice amount, average invoice amount, and logarithmic invoice amount
Supplier activity	Invoice count over 30 days, average amount over 90 days, and supplier age
Supplier risk	Supplier risk score and blacklisted-supplier flag
Amount anomalies	Invoice amount z-score and amount-to-average ratios
Invoice patterns	Duplicate-invoice and split-invoice flags
Submission timing	Submission hour, day of week, weekend flag, and outside-business-hours flag
Budget information	Annual budget and invoice-to-department-budget ratio
OCR and image checks	OCR amount, invoice-OCR difference, metadata availability, and image-tampering flag
Payment terms	NET30, NET60, and NET90 indicators
Invoice type	Goods and services indicators

The variety of features allows the model to evaluate an invoice from several perspectives. For example, the invoice amount may appear reasonable independently but may become suspicious when compared with the supplier's historical average or the department's annual budget.

3.3 Data Preparation and Validation

Data preparation was necessary to ensure that the information provided to the model was complete, correctly structured, and suitable for prediction. The application checks every uploaded file before sending any records to the Databricks endpoint.

The validation process includes:

- Confirming that the uploaded file is in CSV or Parquet format.
- Verifying that the invoice_id column is available.
- Checking that all 32 model features are present.
- Preserving the required feature order.
- Identifying missing or incorrectly structured columns.
- Converting records into a format accepted by the serving endpoint.
- Dividing large datasets into smaller scoring batches.

If required features are missing, the application displays an error instead of sending incomplete records to the model. This reduces the possibility of unreliable predictions and helps users correct the input file.

The invoice identifier is retained for reporting and investigation but is not treated as a predictive feature. After prediction, the identifier relates to the returned fraud probability, score, priority, batch information, and investigation status.



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3.4 Feature Engineering

Feature engineering was used to convert raw invoice information into variables that represent meaningful fraud-risk patterns. The engineered features were designed to capture unusual amounts, suspicious timing, supplier history, invoice duplication, document differences, and payment characteristics.

Important, engineered features include:

- Invoice amount z-score: Measures how far an invoice amount differs from the expected amount pattern.
- Amount-to-supplier-average ratios: Compare the current invoice with the supplier's historical invoice values.
- Amount difference from the 90-day supplier average: Measures the absolute difference between the current invoice and recent supplier activity.
- Supplier age in years: Converts supplier age from days into a more interpretable value.
- Weekend and outside-business-hours flags: Identify invoices submitted during unusual periods.
- Invoice-to-department budget ratio: Measures the invoice value in relation to the available annual budget.
- Logarithmic invoice amount: Reduces the effect of highly skewed invoice values.
- Invoice-OCR differences: Compare the reported invoice value with the value extracted from the document.
- Payment-term indicators: Convert payment-term categories into model-ready binary variables.
- Invoice-type indicators: Represent whether an invoice relates to goods or services.

The final feature set combines original and engineered variables. This gives the model a structured representation of both the invoice itself and its wider supplier, timing, budget, and document context. The completed data-preparation process produced a validated feature structure that could be used consistently during model training and application-based scoring. This consistency is important because the model-serving endpoint must receive the same features and feature definitions that were used when the model was trained.

3.5 Fraud Detection Model Development

A supervised machine learning classification approach was used to estimate the probability that an invoice may be fraudulent. In supervised learning, the model learns patterns from historical records for which the outcome is already known. Each training record contains invoice-related features and a target label indicating whether the transaction was classified as Fraud or Legitimate.

Before model training, the dataset was divided into input features and the target variable. The input data contained the 32 behavioral, supplier, transactional, temporal, financial, and document-related features. The target variable represented the verified fraud outcome.

The model-development process included the following stages:

1. Loading the prepared invoice dataset.
2. Separating the model features from the target label.
3. Dividing the data into training and testing datasets.
4. Training the classification model on the training dataset.
5. Generating predictions for the testing dataset.
6. Evaluating the model using suitable classification measures.
7. Selecting the final model for deployment.



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8. Registering and serving the selected model through Databricks.

The purpose of the model is not simply to produce a Fraud or Legitimate label. It generates a probability representing the estimated fraud risk of each invoice. Probability-based output is more useful for investigation because it allows invoices to be ranked from lower to higher risk.

For example, an invoice with a probability of 0.85 is considered more urgent than an invoice with a probability of 0.35. This ranking allows the application to allocate different investigation priorities instead of treating all predicted cases equally.

3.6 Model Evaluation

The model was evaluated using the testing dataset to determine how effectively it distinguished between fraudulent and legitimate invoices. In fraud detection, model evaluation requires more than overall accuracy because fraudulent transactions may represent a smaller proportion of the data.

The following measures are relevant to model evaluation:

Metric	Purpose
Accuracy	Measures the proportion of all correctly classified invoices
Precision	Measures of how many invoices predicted as fraud were fraudulent
Recall	Measures of how many actual fraudulent invoices were identified
F1-score	Balances precision and recall
ROC-AUC	Measures the model's ability to rank fraudulent invoices above legitimate invoices
Confusion matrix	Summarizes true positives, true negatives, false positives, and false negatives

Recall is particularly important when the objective is to avoid missing fraudulent invoices. However, a model that produces too many false alerts may increase the workload of investigators. Precision is therefore also important. The model should provide a practical balance between detecting suspicious invoices and limiting unnecessary manual reviews.

The final application uses fraud probabilities instead of relying entirely on a single classification threshold. This allows the output to be divided into several priority levels and gives investigators a more flexible method for managing cases.

3.7 Databricks Model Serving

After development and evaluation, the selected model was deployed through Databricks Model Serving. Model serving makes the trained model available through an API endpoint. The Streamlit application can send invoice features to this endpoint and receive predictions without loading or retraining the model inside the web application.

The model-serving workflow is:

Validated Invoice Features → REST API Request → Databricks Model Endpoint → Fraud Probabilities → Streamlit Application



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This separation provides several advantages. The model can be managed independently from the user interface, the Streamlit application remains lightweight, and updated model versions can be deployed through Databricks without rebuilding the complete application.

Authentication credentials, endpoint information, and other sensitive configuration values are stored through Streamlit Secrets. They are not written directly into the application source code or uploaded to the public GitHub repository.

3.8 Batch Prediction Process

RiskCheck AI supports the scoring of multiple invoice files in CSV and Parquet formats. After uploading, the application combines or processes the batches while preserving the invoice identifier and batch information.

Large datasets are divided into smaller chunks before being sent to the serving endpoint. Chunked processing reduces the possibility of request-size errors and allows the application to handle a greater number of invoice records.

Figure 1 shows the validated preview of a demonstration CSV batch before scoring. The interface confirms the uploaded file, displays representative feature columns, reports the number of detected invoices, and enables scoring only after the batch is ready.

The screenshot displays the 'Upload invoice batches' section of the application. A file named 'Test Invoices.csv' (1.4 KB) has been uploaded. Below the upload area, a 'Preview: Test Invoices.csv' table is shown, containing 10 rows of invoice data. The table includes columns for invoice_id, invoice_amount, submission_hour, supplier_invoice_count_30d, supplier_avg_amount_30d, invoice_amount_score, duplicate_invoice_flag, split_invoice_flag, late_submission_flag, supplier_age_days, supplier_risk_score, and blacklisted_flag. Below the table, it states '34 invoice(s) detected.' and '1 file(s) ready for scoring.' A red button labeled 'Score invoice batches' is visible, along with a link to 'Expected upload schema (33 columns)'. The interface also includes a 'Manage app' link in the bottom right corner.

	invoice_id	invoice_amount	submission_hour	supplier_invoice_count_30d	supplier_avg_amount_30d	invoice_amount_score	duplicate_invoice_flag	split_invoice_flag	late_submission_flag	supplier_age_days	supplier_risk_score	blacklisted_flag
0	NBR-2026-09001	420	10	8	460	-0.35	0	0	0	1850	0.08	0
1	NBR-2026-09002	1250	14	12	1180	0.22	0	0	0	2400	0.14	0
2	NBR-2026-09003	3100	19	5	3450	1.15	0	1	0	980	0.42	0
3	NBR-2026-09004	6900	23	2	2800	2.75	0	1	1	420	0.67	0
4	NBR-2026-09005	9800	1	1	2100	3.9	1	1	1	120	0.89	1
5	NBR-2026-09006	560	9	15	610	-0.18	0	0	0	3100	0.05	0
6	NBR-2026-09007	4200	22	3	2300	2.1	0	0	1	640	0.58	0
7	NBR-2026-09008	15000	2	1	3200	4.4	1	0	1	75	0.95	1
8	NBR-2026-09009	2100	11	7	2050	0.12	0	0	0	1500	0.2	0
9	NBR-2026-09010	7600	0	2	2500	3.2	0	1	1	260	0.78	0

Figure 1. Validated invoice-batch preview and scoring control

For every chunk, the application:

1. Selects the 32 required model features.
2. Convert the records into the endpoint's expected request format.
3. Sends the request to Databricks Model Serving.
4. Receives the prediction response.
5. Extracts the fraud probability for each invoice.



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6. Matches each probability with the correct invoice identifier.
7. Adds the result to the complete scored batch.

After all chunks have been processed, the separate results are combined into one scoring output. Each record then contains the original invoice information, fraud probability, RiskCheck Score, investigation priority, and batch identifier.

The completed model-serving process demonstrates how a trained machine learning model can be integrated into an operational application. Instead of remaining within a development notebook, the model becomes available for repeated invoice-risk assessment through a user-friendly interface.

3.9 RiskCheck Score Calculation

The fraud probability returned by the Databricks Model Serving endpoint ranges from 0 to 1. Although this value is suitable for technical analysis, it may be less convenient for accounts payable to users and investigators. RiskCheck AI therefore converts the probability into a score between 0 and 100.

The score is calculated as:

$$\text{RiskCheck Score} = \text{Fraud Probability} \times 100$$

For example, if the model returns a fraud probability of 0.86, the corresponding RiskCheck Score is:

$$0.86 \times 100 = 86$$

A higher score indicates that the invoice contains a stronger combination of patterns associated with fraud. A lower score indicates that the transaction is closer to the patterns generally observed in legitimate invoices.

The RiskCheck Score does not represent the percentage certainty that fraud has occurred. It is a prioritization measure derived from the model probability. Investigators should use it together with invoice details, supplier information, supporting documents, and the organization's internal policies.

3.10 Investigation Priority Classification

RiskCheck AI groups invoices into five priority levels based on their scores. This provides a consistent method for organizing the investigation queue.

Priority	Score range	Risk level	Recommended action
Priority 1	80–100	Critical	Immediate fraud investigation
Priority 2	60–79.99	High	High-priority manual review
Priority 3	40–59.99	Elevated	Standard manual review
Priority 4	20–39.99	Moderate	Monitoring or secondary review
Priority 5	0–19.99	Low	Routine processing



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Priority 1 – Critical Risk

Priority 1 includes invoices with scores of 80 or above. These invoices contain the strongest risk patterns identified by the model and require immediate investigation. Reviewers should examine the supplier, transaction amount, invoice document, OCR results, duplication indicators, timing information, and other relevant evidence before approving payment.

Priority 2 – High Risk

Priority 2 includes invoices with scores from 60 to below 80. These invoices require urgent manual review but are placed below critical cases in the investigation queue. Additional verification may include confirming supplier details, matching the invoice with a purchase order, and checking historical transactions.

Priority 3 – Elevated Risk

Priority 3 includes invoices with scores from 40 to below 60. These invoices require a standard manual review. They may contain unusual characteristics, but the overall combination of features is less severe than that of Priority 1 and Priority 2 cases.

Priority 4 – Moderate Risk

Priority 4 includes invoices with scores from 20 to below 40. These records may continue through additional monitoring or secondary checks according to the organization's policies and available investigation capacity.

Priority 5 – Low Risk

Priority 5 includes invoices with scores below 20. These invoices may continue through routine processing, subject to existing financial controls. A low score does not guarantee that an invoice is legitimate; it only indicates that the model identified fewer fraud-related patterns.

3.11 Investigation Prioritization Results

The priority framework converts model output into an operational investigation queue. Instead of presenting investigators with an unsorted list of transactions, the system places the most suspicious invoices at the top. Figure 2 presents the scored results for the demonstration batch. Summary indicators and risk-distribution charts allow the reviewer to assess the overall batch and compare invoice scores before opening individual cases.



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Figure 2. Batch-level risk summary and invoice-score distribution

For every scored invoice, the application displays:

- Invoice identifier
- Fraud probability
- RiskCheck Score
- Priority level
- Risk description
- Recommended investigation action
- Batch identifier

This structure helps reviewers understand which invoices require attention and why they have been prioritized. It also provides a consistent approach across different invoice batches and investigators.

Figure 3 shows the operational priority queue. Search and filter controls help investigators locate invoices and focus on specific risk categories or priority levels, while the table records the score, recommended action, and investigation reasons.

Search Invoice		Risk category		Priority	
Enter invoice ID...		All		All	
Invoice ID	RiskCheck Score	Risk Category	Priority Level	Recommended Action	Investigation Reasons
NBR-2026-09020	98.95	Critical Risk	Priority 1 - Immediate	Immediate fraud investigation	Invoice amount differs substantially from recent supplier pattern Large deviation from supplier
NBR-2026-09027	96.27	Critical Risk	Priority 1 - Immediate	Immediate fraud investigation	Supplier is blacklisted Invoice amount differs substantially from recent supplier pattern Large
NBR-2026-09015	94.33	Critical Risk	Priority 1 - Immediate	Immediate fraud investigation	Invoice amount differs substantially from recent supplier pattern Large deviation from supplier
NBR-2026-09034	94.66	Critical Risk	Priority 1 - Immediate	Immediate fraud investigation	Split-invoice behaviour detected Invoice amount differs substantially from recent supplier patts
NBR-2026-09018	93.56	Critical Risk	Priority 1 - Immediate	Immediate fraud investigation	Split-invoice behaviour detected Invoice amount differs substantially from recent supplier patts
NBR-2026-09025	91.61	Critical Risk	Priority 1 - Immediate	Immediate fraud investigation	Split-invoice behaviour detected Invoice amount differs substantially from recent supplier patts
NBR-2026-09022	90.86	Critical Risk	Priority 1 - Immediate	Immediate fraud investigation	Split-invoice behaviour detected Invoice amount differs substantially from recent supplier patts
NBR-2026-09030	90.28	Critical Risk	Priority 1 - Immediate	Immediate fraud investigation	Split-invoice behaviour detected Invoice amount differs substantially from recent supplier patts
NBR-2026-09013	90.03	Critical Risk	Priority 1 - Immediate	Immediate fraud investigation	Split-invoice behaviour detected Supplier is blacklisted Invoice amount differs substantially
NBR-2026-09005	87.87	Critical Risk	Priority 1 - Immediate	Immediate fraud investigation	Split-invoice behaviour detected Supplier is blacklisted Invoice amount differs substantially

Showing 34 of 34 invoices in this batch.

Figure 3. Searchable and filterable investigation priority queue



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Figure 4 demonstrates the individual investigation view. It brings together the RiskCheck Score, risk category, urgency, model-supported reasons, and recommended action so that the reviewer can examine the case in context.

Figure 4. Individual invoice investigation view

3.12 Automated Priority Email Alerts

RiskCheck AI includes an automated email-alert process for Priority 1 and Priority 2 invoices. When a batch contains critical-risk or high-risk records, the application prepares a formatted HTML email for the configured recipients.

The email contains individual risk cards showing important information about each flagged invoice. It also includes a direct application link containing the relevant batch identifier. When a reviewer follows the link, the application can open the associated batch for investigation.

Different recipient lists can be configured for Priority 1 and Priority 2 alerts. This allows critical invoices to be sent to senior fraud investigators or escalation teams while high-risk invoices are directed to the appropriate accounts payable reviewers.

The alert workflow can be represented as:

High-Risk Invoice Detected → Email Generated → Reviewer Notified → Batch Opened → Invoice Investigated

Email notifications reduce the time between risk detection and manual review. They also help ensure that critical cases are not overlooked when several batches are processed.

The notification process does not confirm that fraud has occurred. Its purpose is to alert authorised reviewers that an invoice has received a high score and requires further investigation. The final outcome is recorded only after the reviewer has examined the available evidence.



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Figure 5 illustrates the reviewer-facing alert format for high-priority invoices. Each risk card includes the invoice identifier, score, category, reasons for the flag, recommended action, and a direct link back to RiskCheck AI.

High Priority Review — P2 (2)

Invoice NBR-2026-09008
 RiskCheck Score
 Risk Category

78.01
High Risk

Why flagged
 Supplier is blacklisted, Invoice amount differs substantially from recent supplier pattern, Large deviation from supplier 90-day average, Elevated supplier risk score, Unusual invoice amount relative to historical distribution

Recommended action
 High-priority manual review

Invoice NBR-2026-09010
 RiskCheck Score
 Risk Category

73.51
High Risk

Why flagged
 Split-invoice behaviour detected, Invoice amount differs substantially from recent supplier pattern, Large deviation from supplier 90-day average, Elevated supplier risk score, Unusual invoice amount relative to historical distribution

Recommended action
 High-priority manual review

[Open RiskCheck AI](#)

RiskCheck AI provides decision-support risk indicators. These alerts are not final fraud determinations. Investigator review is required before any final decision or action.

Figure 5. Automated high-priority investigation email alert

3.13 Streamlit Application

A Streamlit application was developed to provide an interactive interface for RiskCheck AI. The application connects the invoice upload process, schema validation, Databricks model predictions, risk prioritization, investigation review, and human feedback within one system.

The application allows accounts payable users and investigators to perform the following activities:

- Upload multiple CSV or Parquet invoice files.
- Validate the required invoice schema.
- Submit invoice records for fraud-risk prediction.
- View fraud probabilities and RiskCheck Scores.
- Filter invoices according to investigation priority.
- Review individual invoice and supplier risk indicators.
- Open a specific batch through an email deep link.
- Record verified Fraud or Legitimate outcomes.
- Monitor whether enough new labels are available for retraining.
- Download or review scored results.



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Streamlit was selected because it supports the rapid development of interactive data applications using Python. It also allows analytical functions, API requests, tables, filters, and investigation controls to be combined in a web-based interface.

The application is deployed through Streamlit Community Cloud. Sensitive values, including Databricks credentials and email passwords, are stored using Streamlit Secrets and are not included directly in the public source code.

3.14 Delta Lake Storage and Audit Trail

All scored invoice records are stored in a Databricks Delta table. Persistent storage is important because model predictions should remain available after the application session has ended.

The scored-invoice table stores information such as:

- Invoice identifier
- Batch identifier
- Fraud probability
- RiskCheck Score
- Investigation priority
- Relevant invoice information
- Scoring and processing details

The default scored-invoice table used by the project is:

`workspace.default.riskcheck_scored_invoices`

Storing the results creates an audit trail of the invoices that were processed and the risk values assigned by the model. This information can support investigation, reporting, monitoring, and later comparison with verified outcomes.

Databricks SQL is used to write and retrieve information from the Delta tables. Large queries are divided into smaller parameter batches when necessary to remain within SQL parameter limits.

3.15 Investigator Feedback Collection

Machine learning predictions are not treated as verified fraud decisions. Investigators review the invoices scored and record the outcome as either Fraud or Legitimate.

The verified feedback is stored in a separate Delta table:

`workspace.default.riskcheck_investigator_feedback`

Separating predictions from human decisions provides clear traceability. It prevents a model-generated risk score from being confused with a confirmed investigation result.

The feedback process can be represented as:

Scored Invoice → Investigator Review → Fraud or Legitimate Outcome → Feedback Storage

Investigator feedback serves two purposes. First, it provides a record of the final review decision. Second, it creates labelled data that can be used to evaluate and improve the fraud detection model.

To maintain feedback quality, outcomes should be entered only after the reviewer has examined the invoice, supplier information, relevant documents, and supporting evidence.



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Figure 6 shows a verified Fraud outcome being saved with an investigator note. The confirmation message also reports progress towards the configured retraining threshold, linking human review directly to controlled model improvement.

Figure 6. Saved investigator feedback and retraining-threshold progress

3.16 Automated Model Retraining

Fraud patterns may change over time. A model trained on historical data may become less effective if supplier behavior, invoice processes, or fraud techniques change. RiskCheck AI therefore includes a feedback-based retraining process.

The application counts the number of verified labels that have not yet been used for model retraining. When this count reaches a configured minimum threshold, such as 200 new labels, the application can trigger a Databricks retraining job through the REST API.

The retraining workflow is:

1. Investigators verify invoice outcomes.
2. Feedback records are stored in Delta Lake.
3. The system counts unconsumed verified labels.
4. The count is compared with the retraining threshold.
5. If the threshold is reached, a Databricks Job is triggered.
6. The retraining notebook prepares the updated dataset.
7. A new model is trained and evaluated.
8. The new model can be reviewed before being used for scoring.

This process supports continuous model improvement while preventing retraining from being triggered after every individual review. The threshold ensures that a meaningful amount of new information is available.



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3.17 System Architecture

RiskCheck AI combines user interaction, model serving, persistent storage, automated communication, and model lifecycle management.

The main system architecture is:

Figure 7 summarises the complete technical and operational architecture, from invoice upload and model serving to Delta storage, reviewer alerts, verified feedback, and threshold-based retraining.

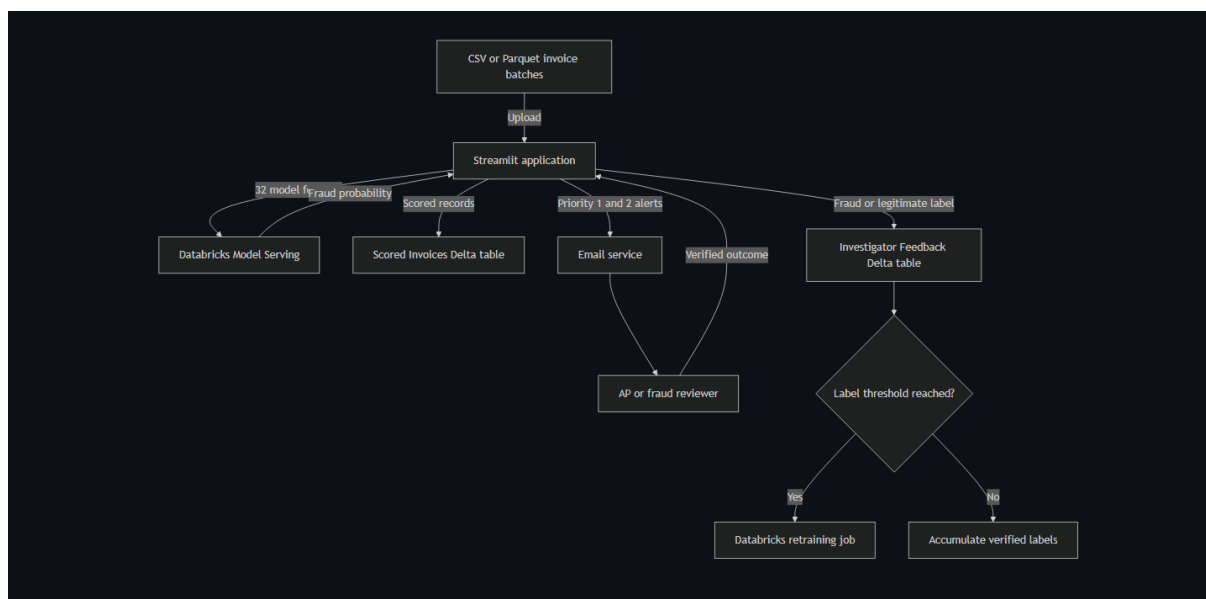


Figure 7. RiskCheck AI system architecture and feedback loop

Invoice Files → Streamlit Application → Databricks Model Serving → Risk Scores and Priorities

Risk Scores → Delta Lake Storage → Investigator Review → Verified Feedback

High-Risk Results → Email Alert → Fraud or AP Reviewer

Verified Feedback → Threshold Check → Databricks Retraining Job

The Streamlit application acts as the main user interface and coordinates the workflow. Databricks Model Serving produces fraud probabilities. Databricks SQL and Delta Lake preserve scored records and verified outcomes. The email service notifies reviewers about urgent invoices, while Databricks Jobs supports automated retraining.

This architecture transforms a machine learning model into a complete investigation-support pipeline. It connects prediction with operational action, human judgement, auditability, and continuous improvement.



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Social & Industry Relevance

4.1 Industry Relevance

Invoice fraud and payment errors are important concerns for organizations that process large volumes of supplier invoices. Duplicate payments, manipulated documents, false suppliers, unusual transaction

amounts, and unauthorized changes can create financial losses and increase the workload of accounts payable teams.

RiskCheck AI demonstrates how data analytics and machine learning can support invoice-review processes. The system evaluates multiple invoice and supplier characteristics, calculates a fraud-risk score, and places invoices into an investigation queue. This risk-based approach can help organizations use their investigation resources more efficiently.

The project is relevant to industries such as retail, manufacturing, logistics, healthcare, financial services, construction, and government. These sectors regularly manage supplier payments and may need to review large numbers of invoices from different vendors, departments, and locations.

4.2 Business Relevance

RiskCheck AI connects machine learning predictions with practical business actions. The application does not stop after identifying a fraud probability. It converts the probability into a RiskCheck Score, assigns an investigation priority, sends alerts for urgent cases, stores results, and records verified outcomes.

The business workflow can be represented as:

Invoice Submission → Fraud-Risk Assessment → Investigation Priority → Human Review → Verified Decision

This workflow provides several possible business benefits:

- Reduces the time required to identify high-risk invoices.
- Creates a consistent approach to investigation prioritization.
- Helps prevent duplicate or suspicious payments.
- Allow reviewers to focus on transactions with the highest estimated risk.
- Preserves prediction and review information for auditing.
- Improves communication through automated alerts.
- Creates verified data for model monitoring and retraining.

The project may also help reduce unnecessary delays for low-risk invoices. Instead of applying the same intensive review to every transaction, the organization can use the risk score to determine the appropriate level of attention.



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4.3 Decision-Support Value

RiskCheck AI is designed as a decision-support system. It does not automatically label a supplier as fraudulent, reject an invoice, or block a payment. The model identifies patterns associated with risk and recommends an investigation priority.

The final decision remains with an authorised reviewer who can examine additional evidence such as purchase orders, delivery records, supplier contracts, approval history, and invoice documents.

Keeping a human reviewer involved is important because model predictions may contain false positives and false negatives. Human judgement is also required to understand the business context surrounding an unusual transaction.

4.4 Social and Ethical Relevance

Fraud detection systems must be used responsibly because incorrect predictions may affect suppliers, employees, and financial operations. RiskCheck AI separates model predictions from verified investigator decisions so that a risk score is not presented as confirmed fraud.

Responsible use of the system requires:

- Protect confidential invoices and supplier information.
- Restricting access to authorised users.
- Monitoring the accuracy and reliability of the model.
- Reviewing high-risk predictions before acting.
- Maintaining clear records of model and human decisions.
- Checking retrained models before deployment.
- Avoiding unfair treatment based only on automated output.

The project therefore demonstrates how automation can support financial control while maintaining human accountability.

Learning & Reflection

5.1 Technical Learning

During the development of RiskCheck AI, I gained practical experience in Python, data preparation, machine learning, Streamlit, Databricks Model Serving, Databricks SQL, Delta Lake, REST APIs, and email automation. I learned how to prepare model features, validate uploaded files, process large datasets in chunks, call an external model endpoint, interpret API responses, and present predictions through an interactive application. I also learned how to store scored records and human feedback in Delta tables.



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The retraining component helped me understand that a machine learning project continues after the initial model has been trained. Models require monitoring, new verified data, evaluation, and controlled updates.

5.2 Analytical and Problem-Solving Learning

The project improved my ability to convert a business problem into an analytical workflow. I learned to identify the needs of accounts payable users, define invoice-risk factors, create a scoring structure, and connect model predictions with recommended actions.

I also developed problem-solving skills while handling data validation, API payload limits, SQL parameter boundaries, application state, email configuration, batch identifiers, and retraining conditions. The development process requires testing individual components and understanding how they interact. A problem in one part of the system, such as an incorrectly formatted feature, could affect the model endpoint, application output, and stored results.

5.3 Internship Experience

The internship provided practical exposure to the complete data analytics process, including data preparation, SQL, Python, exploratory analysis, feature engineering, machine learning, business intelligence, and application development.

RiskCheck AI allowed me to combine these skills into one capstone project. It improved my confidence in developing an application independently and helped me understand the importance of documentation, testing, security, responsible machine learning, and clear communication.

Overall, the project strengthened both my technical knowledge and my ability to present analytical results in a form that supports real business decisions.

Future Scope & Conclusion

6.1 Future Scope

RiskCheck AI demonstrates complete invoice fraud detection and investigation workflow. However, the project can be expanded to improve its accuracy, security, scalability, and suitability for production environments.

Integration with Live Financial Systems

The application can be connected to enterprise resource planning and accounting systems. Instead of requiring users to upload files manually, invoices could be received directly from procurement or accounts payable platforms. This would allow risk scoring to become part of the organization's normal invoice-processing workflow.



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Real-Time Invoice Scoring

Future versions could score invoices immediately after they are submitted. Real-time scoring would allow suspicious transactions to be identified before approval or payment, reducing the delay between invoice submission and investigation.

Explainable Artificial Intelligence

An explainability component could be added to show the factors that contributed most strongly to each RiskCheck Score. For example, the application could indicate that an invoice received a high score because of an unusual amount, a blacklisted supplier, a duplicate-invoice indicator, or a large OCR difference. These explanations would help investigators understand model predictions and improve confidence in the review process.

Advanced Document Analysis

The document-processing component could be extended using optical character recognition, computer vision, and document-layout analysis. Future versions could examine invoice logos, signatures, bank details, fonts, metadata, and formatting inconsistencies. The system could also compare invoices with previous documents submitted by the same supplier to identify unexpected changes.

Supplier Network Analysis

Graph-based analysis could be used to identify relationships between suppliers, bank accounts, addresses, employees, and transactions. This may help detect organized fraud patterns that are difficult to identify by reviewing invoices separately.

Role-Based Access Control

A production version should include different access levels for accounts payable by users, investigators, managers, auditors, and system administrators. Each user should only be able to access the information and actions required for their role.

Model Monitoring and Drift Detection

The system could monitor changes in feature distributions, fraud rates, prediction confidence, and investigation outcomes. This would help identify model drift and determine when retraining is necessary. A monitoring dashboard could display model performance, false-alert rates, feedback volumes, retraining history, and the performance of different model versions.

Improved Retraining Governance

The automated retraining workflow could include formal approval stages. A newly trained model should be evaluated against the current production model before deployment. Only models that meet defined performance and fairness requirements should be promoted.

Model versioning and rollback support could also be added so that an earlier model can be restored if a new version performs poorly.



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Additional Notification Channels

Future versions could integrate with communication and case-management platforms such as Microsoft Teams, Slack, Jira, or ServiceNow. High-risk invoices could automatically create investigation cases and assign them to the appropriate reviewer.

Investigation Case Management

The application could be extended with investigation notes, evidence attachments, reviewer assignments, escalation status, and approval history. This would transform RiskCheck AI from a scoring application into a more complete fraud case-management system.

6.2 Conclusion

RiskCheck AI was developed as an end-to-end invoice fraud detection and investigation-prioritization system. The project combines data preparation, feature engineering, machine learning, model serving, risk scoring, automated email alerts, Delta Lake storage, human feedback, and automated retraining in one workflow. The system processes CSV and Parquet invoice batches and validates the 32 features required by the fraud detection model. Databricks Model Serving generates a fraud probability for every invoice. This probability is converted into a RiskCheck Score and assigned to one of five investigation priorities. Critical-risk and high-risk invoices can generate automated alerts, allowing reviewers to respond quickly. Scored invoices are stored in Delta Lake to maintain an audit trail, while verified Fraud and Legitimate outcomes are stored in a separate feedback table. When enough new verified labels are available, the application can trigger a Databricks retraining job.

The project successfully connects machine learning output with practical accounts payable actions. It demonstrates that an effective fraud detection solution requires more than a predictive model. It also requires reliable data validation, understandable risk levels, communication with reviewers, persistent records, human verification, and a controlled model-improvement process. RiskCheck AI remains a decision-support system. A high score identifies an invoice that requires investigation but does not confirm fraud. The final decision remains with an authorized investigator who can examine the complete business context and supporting evidence.

Developing this project allowed me to apply the knowledge and skills gained during my Data Analytics Internship and Capstone Program. It improved my understanding of Python, data analysis, machine learning, Databricks, SQL, Delta Lake, APIs, Streamlit application development, automation, and responsible artificial intelligence. Overall, RiskCheck AI demonstrates how data analytics and machine learning can support faster, more consistent, and more accountable invoice fraud investigations.



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